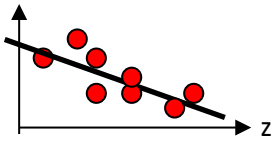
**Statistics**
**Lecture 09**
**Linear regression: models**

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**Simple linear regression**

Model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$

Fit of 1-dimensional regression line
to 2-dimensional observations (x,Y)




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**Pairwise data**
**Tasks for linear regression**
**Pre-analysis**
**Simple linear regression model**
**Assumptions**
**Cost function**
**Optimization**
**Estimate of error variance**
**Model check**
**Sum of squares decomposition and model fit measure $R^2$**

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**Demo**

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**The concept of linear regression**
**Non-linear regression transformable to linear regression**
**Non-linear regression NOT transformable to linear regression**
**(Multiple linear regression)**

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